

The empirical recurrence for OEIS A236879: recovering a conjecture that is not written down

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Abstract

OEIS A236879 records “Empirical recurrence of order 63”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 251$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236879 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum plus the upper median plus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

537, 2111, 7900, 40199, 189527, 1031110, 5389465, 30143115, ...

The entry states:

Empirical recurrence of order 63 (see link above).

As of the “Last modified” line on the live entry (Nov 05 2014 17:54:52, revision 9) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 251$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 251$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 502$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	16	c_{17}	−806626246425	c_{33}	−145011176324599384	c_{49}	−47417347574292480
c_2	27	c_{18}	−5549283296459	c_{34}	−1055842701605466	c_{50}	22079229293202432
c_3	−1667	c_{19}	8679696177536	c_{35}	259241512822452234	c_{51}	14817525871303680
c_4	4072	c_{20}	34723994107362	c_{36}	−34025039397003762	c_{52}	−6514237146490880
c_5	76934	c_{21}	−68113932764134	c_{37}	−381021968390179388	c_{53}	−3524453417234432
c_6	−341187	c_{22}	−174240406972383	c_{38}	91434397011877932	c_{54}	1390125346611200
c_7	−2049421	c_{23}	414218469332205	c_{39}	458967239587347048	c_{55}	620930322202624
c_8	13078549	c_{24}	701207183888323	c_{40}	−148931424202336496	c_{56}	−205596162195456
c_9	33627314	c_{25}	−2008642190732129	c_{41}	−451028233509798832	c_{57}	−77774770995200
c_{10}	−318556323	c_{26}	−2246830446293595	c_{42}	175195326671834064	c_{58}	19696234659840
c_{11}	−306610231	c_{27}	7889578094693248	c_{43}	359397679149116544	c_{59}	6478165180416
c_{12}	5496429956	c_{28}	5619591657864136	c_{44}	−155964021652045664	c_{60}	−1082279854080
c_{13}	54510153	c_{29}	−25328063997365173	c_{45}	−230443907502100160	c_{61}	−317278126080
c_{14}	−70935843916	c_{30}	−10441277176211870	c_{46}	106480622832011904	c_{62}	25480396800
c_{15}	45538376655	c_{31}	66791970202879633	c_{47}	117774941203385216	c_{63}	6794772480
c_{16}	706944602204	c_{32}	12296932132110369	c_{48}	−55687378365970432		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63 + 4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

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